Chapter 2 · Linear Relationships

Algebra 1 | Practice | Two questions per page

01 Function rule

Functions, Domain & Range

Use y = 4x - 1 to find the outputs for x = 0, 2, 4, and 6.

Work space

02 Function / domain / range

Functions, Domain & Range

For {(-2, 1), (1, 5), (-2, 3), (4, 9)}, decide whether the relation is a function. List its domain and range.

Work space

Show work where needed. Include units for rates of change.

03 Same output

Functions, Domain & Range

Write two ordered pairs with the same output of 7 but different inputs. Is your relation a function?

Work space

04 Input-output table

Functions, Domain & Range

A game costs $7 to enter plus $4 per round. Complete a table for 0, 1, 2, and 3 rounds. List the domain and range.

Work space

05 Solve an equation

Functions, Domain & Range

Solve 5(x - 2) = 3x + 8. Show your work.

Work space

06 Constant change

Functions, Domain & Range

The inputs are 0, 1, 2, 3 and the outputs are 11, 7, 3, -1. Find the change in output for each increase of 1 in the input.

Work space

07 Slope from a table

Tables, Graphs & Constant Change

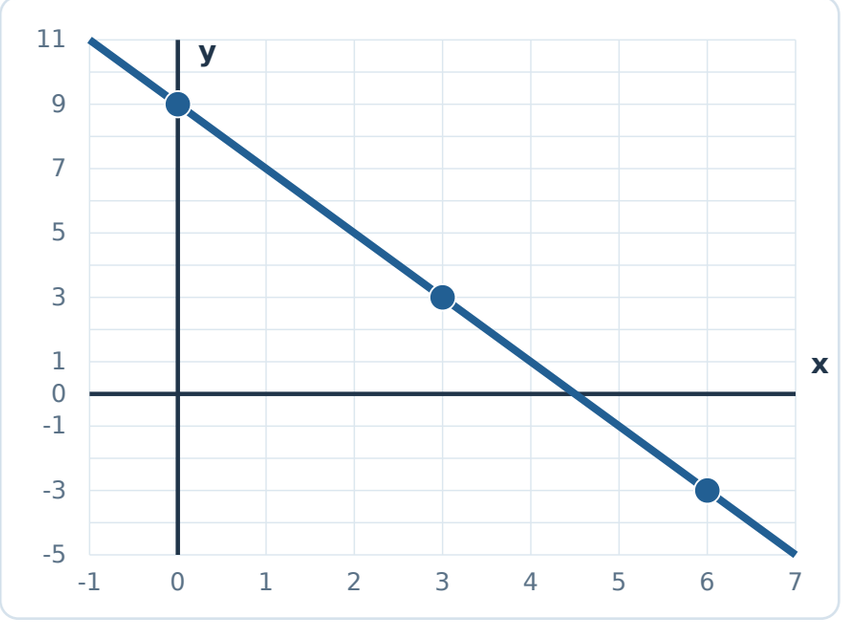
A table has x-values 0, 2, 4, 6 and y-values 4, 10, 16, 22. Find the slope. Show Δy and Δx.

Work space

08 Slope from a graph

Tables, Graphs & Constant Change

Find the slope and starting value of the line shown.



Work space

09 Write a linear rule

Tables, Graphs & Constant Change

The points (0, -2), (1, 1), (2, 4), (3, 7) form a linear relationship. Find the slope and write y = mx + b.

Work space

10 Compare steepness

Tables, Graphs & Constant Change

Pattern A increases its output by 3 for every increase of 1 in the input. Pattern B increases by 6. Which graph is steeper?

Work space

11 Repeated inputs

Tables, Graphs & Constant Change

Is {(1, 4), (1, 4), (4, 7)} a function? List the domain and range.

Work space

12 Rate with units

Tables, Graphs & Constant Change

A runner changes position by 28 meters in 4 seconds. Find the average rate of change, including units.

Work space

13 Signed rate

Slope, Motion & Comparing Rates

An object is at 35 m at 0 s and 11 m at 8 s. Find the average rate of change with units. Is its position increasing or decreasing?

Work space

14 Slope from two points

Slope, Motion & Comparing Rates

Find the slope between (-4, 7) and (2, -11). Show Δy and Δx.

Work space

15 Compare rates

Slope, Motion & Comparing Rates

Cyclist A travels 40 m in 5 s. Cyclist B travels 54 m in 9 s. Find both average rates and identify the faster cyclist.

Work space

16 Decreasing quantity

Slope, Motion & Comparing Rates

A tank begins with 82 gallons and loses 6 gallons each minute. Write an equation for the amount after t minutes. Identify the starting value and rate.

Work space

17 Function check

Slope, Motion & Comparing Rates

Is {(0, 6), (3, 6), (3, 8)} a function? List the domain and range.

Work space

18 Write a motion rule

Slope, Motion & Comparing Rates

At t = 0, a scooter is 9 meters from a marker. Its position increases by 5 meters each second. Write an equation for position p after t seconds.

Work space

19 Cost model

Writing & Interpreting Linear Models

A membership costs $8 per month plus $3 per movie. Write an equation for total monthly cost C for n movies. Identify the starting fee and rate.

Work space

20 Table to equation

Writing & Interpreting Linear Models

A table contains (0, 18), (1, 14), (2, 10), (3, 6). Write its equation in y = mx + b form.

Work space

21 Equation from two points

Writing & Interpreting Linear Models

Write the equation of the line through (0, -3) and (4, 9). Show m and b.

Work space

22 Match a rule

Writing & Interpreting Linear Models

A relationship gives outputs 6, 11, 16 for inputs 0, 1, 2. Write the linear equation.

Work space

23 Interpret a model

Writing & Interpreting Linear Models

Position is p = 26 - 4t (meters, seconds). State the starting position and rate of change. Find p when t = 5.

Work space

24 Function and equation

Writing & Interpreting Linear Models

Is {(0, 5), (1, 8), (2, 11), (2, 11)} a function? Write a linear rule that fits the points.

Work space

25 Slope and a point

Optional: More Equation Practice

A line has slope -2 and passes through (3, 4). Write its equation in y = mx + b form.

Work space

26 Two points and a check

Optional: More Equation Practice

Find the equation of the line through (-1, 9) and (3, -3). Does (2, 0) lie on the line?

Work space